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REVIEWER ACTIVITIES: NAVY-AS, DLA-IS

USER ACTIVITIES: NAVY-MC

DD FORM 672-1 (COORDINATED)

PREPARING ACTIVITY: NAVY-OS (Other Cust)

AGENCY: ARMY-AT, AIR FORCE-11

PROCUREMENT SPECIFICATION: MIL-W-21338

INTERNATIONAL INTEREST:

SUPERSEDES:

TITLE: WASHER, KEY RETAINING, BALL AND ROLLER BEARING, REGULAR SERIES

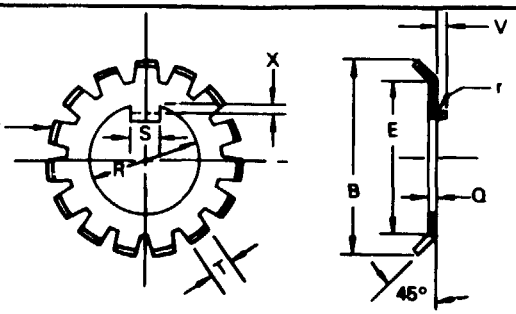
MILITARY STANDARD

MS19070

SHEET 1 OF 2

PROPOSED REVISION 5310-0948

FSC: 5310 (A)



(A) TANGS EQUALLY SPACED AND EACH TANG LOCATED WITHIN .010 INCH OF TRUE POSITION

DASH NO.	Q THICKNESS		R BORE		E DIAMETER		B DIA OVER TANGS	KEY							TANGS			(A)
								S WIDTH		X DEPTH		V PROJECTION		r RAD	NO.	T WIDTH		
	Q	TOL ±	MAX	MIN	E	TOL +	MAX	MAX	MIN	MAX	MIN	V	TOL +			T	TOL ±	
00	.042	.004	.421	.406	0.625	.015	0.875	.120	.110	.030	.020	.062	.031	.016	9	.115	.005	
01	.042	.004	.499	.484	0.719	.015	1.016	.120	.110	.030	.020	.062	.031	.016	9	.115	.005	
02	.042	.004	.616	.601	0.812	.015	1.156	.120	.110	.030	.020	.062	.031	.016	11	.115	.005	
03	.042	.004	.694	.679	0.938	.015	1.328	.120	.110	.030	.020	.062	.031	.016	11	.115	.005	
04	.042	.004	.816	.801	1.125	.015	1.531	.176	.156	.030	.020	.062	.031	.016	11	.156	.010	
05	.050	.004	1.009	.989	1.281	.015	1.719	.176	.156	.030	.020	.094	.031	.016	13	.156	.010	
06	.050	.004	1.213	1.193	1.500	.015	1.922	.176	.156	.050	.035	.094	.031	.031	13	.156	.010	
*065	.050	.004	1.353	1.333	1.812	.015	2.250	.176	.156	.050	.035	.094	.031	.031	15	.156	.010	
07	.050	.004	1.416	1.396	1.812	.015	2.250	.176	.156	.050	.035	.094	.031	.031	15	.156	.010	
08	.058	.005	1.603	1.583	2.000	.030	2.469	.290	.250	.050	.035	.094	.031	.031	15	.219	.015	
09	.058	.005	1.817	1.792	2.281	.030	2.734	.290	.250	.050	.035	.125	.031	.031	17	.219	.015	
10	.058	.005	2.017	1.992	2.438	.030	2.922	.290	.250	.050	.035	.125	.031	.031	17	.219	.015	
11	.063	.005	2.207	2.182	2.656	.030	3.109	.290	.250	.050	.035	.125	.031	.031	17	.219	.015	
12	.063	.005	2.425	2.400	2.844	.030	3.344	.290	.250	.070	.055	.125	.031	.047	17	.219	.015	
13	.063	.005	2.613	2.588	3.062	.030	3.578	.290	.250	.070	.055	.125	.031	.047	19	.219	.015	
14	.063	.005	2.816	2.791	3.312	.030	3.828	.290	.250	.070	.055	.188	.062	.047	19	.219	.015	
15	.072	.005	3.003	2.973	3.562	.030	4.109	.290	.250	.070	.055	.188	.062	.047	19	.313	.015	
16	.072	.005	3.207	3.177	3.844	.030	4.375	.353	.313	.070	.055	.188	.062	.047	19	.313	.015	
17	.072	.005	3.425	3.395	4.031	.030	4.625	.353	.313	.070	.055	.188	.062	.047	19	.313	.015	
18	.094	.006	3.612	3.582	4.281	.045	4.938	.353	.313	.070	.055	.188	.062	.047	19	.313	.015	
19	.094	.006	3.830	3.800	4.562	.045	5.219	.353	.313	.070	.055	.188	.062	.047	19	.313	.015	
20	.094	.006	4.018	3.988	4.812	.045	5.500	.353	.313	.085	.065	.250	.062	.047	19	.313	.015	
21	.094	.006	4.222	4.192	5.000	.045	5.703	.353	.313	.085	.065	.250	.062	.047	19	.375	.015	
22	.125	.007	4.425	4.395	5.281	.045	6.062	.353	.313	.085	.065	.250	.062	.047	19	.375	.015	
24	.125	.007	4.831	4.801	5.688	.045	6.469	.353	.313	.085	.065	.250	.062	.047	19	.375	.015	
26	.125	.007	5.226	5.191	6.188	.045	7.031	.435	.375	.105	.085	.250	.062	.062	19	.500	.020	
28	.125	.007	5.617	5.582	6.531	.045	7.438	.590	.500	.105	.085	.250	.062	.062	19	.500	.020	
30	.156	.008	6.018	5.983	7.062	.060	8.062	.590	.500	.105	.085	.312	.062	.062	19	.500	.020	
32	.156	.008	6.424	6.389	7.438	.060	8.438	.590	.500	.105	.085	.312	.062	.062	19	.500	.020	
34	.156	.008	6.799	6.764	8.031	.060	9.062	.715	.625	.105	.085	.312	.062	.062	19	.500	.020	
36	.156	.008	7.206	7.171	8.375	.060	9.438	.715	.625	.105	.085	.312	.062	.062	19	.625	.020	
38	.156	.008	7.612	7.577	8.781	.060	9.875	.715	.625	.105	.085	.312	.062	.062	19	.625	.020	
40	.156	.008	8.017	7.982	9.156	.060	10.312	.840	.750	.105	.085	.312	.062	.062	19	.625	.020	

(A) *Inactive for new design after 1987

For NOTES see sheet 2. (A) denotes changes.

APPROVED 7 JUN 1961 REVISED (A) 7 October 1987

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NOTES:

FED. SUP CLASS 5310 (A)

(A) 1. MATERIALS:
(a) Carbon steel, AISI 1010 to 1035 (UNS G10100 to G10350), or 1116 to 1118 (UNS G11160 to G11180).
(b) Corrosion resisting steel AISI 302, 303 or 304 (UNS S30200, S30300, or S30400).

2. TEMPER: Each washer shall be of such temper as to permit its being bent 180°, without fracture, either way of the grain, over a radius equal to one half the stock thickness.

(A) 3. PROTECTIVE COATING:
(a) Carbon steel washers are available either zinc plated or unplated. Zinc plated washers shall be plated in accordance with ASTM B633-78, Class 5, Type II. (Cadmium plating in accordance with QQ-P-416, Type II, Class 3 may no longer be furnished.
(b) Corrosion resisting steel washers shall be passivated in accordance with QQ-P-35, Type I, II, or III.

4. DIMENSIONS: All dimensions are in inches. Tolerances: decimals ± .010, angles ± 2° unless otherwise specified.

(A) 5. PART NUMBER: The MS part number is the MS number, plus the dash and the material number. The material numbers are: 1—unplated carbon steel
2—zinc plated carbon steel
3—corrosion resisting steel

Examples: MS19070-002
- material number
- dash number

6. NUTS: See MS19068 for applicable nuts. Applicable nuts should have size numbers corresponding to washer dash numbers: e.g., washer size 00 should be used with nut size number 00.

7. OTHER NOTES:
(a) In the event of a conflict between the text of this standard and the references cited herein, the text of this standard shall take precedence.

(b) Referenced Government documents of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation form a part of this standard to the extent specified herein.

APPROVED 7 JUN 61 REVISED (A) For changes see sheets 1 and 2.

USER ACTIVITIES:

REVIEWER ACTIVITIES:

This military standard is approved for use by all Departments and Agencies of the Department of Defense. Selection for all new engineering and design applications and for repetitive use shall be made from this document.

P. A.	INTERNATIONAL INTEREST	TITLE	MILITARY STANDARD
NAVY-OS Other Cust ARMY-AT AIR FORCE-11		WASHER, KEY RETAINING, BALL AND ROLLER BEARING, REGULAR SERIES	MS19070
PROCUREMENT SPECIFICATION MIL-W-21338	SUPERSEDES:		SHEET 2 OF 2

DD FORM 672-1 COORDINATED

PROPOSED REVISION 5310-0948

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